

IsDB-BISEW

Project Proposal

Student Management System (PHP & MySQL)

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1. Project Overview

The Student Management System (SMS) is a web-based application developed using PHP and MySQL that enables educational institutions to efficiently manage student information, attendance, marks, fees, and academic records. It replaces manual record-keeping with a centralized, secure, and user-friendly digital platform.

2. Project Objectives

- Digitize and centralize student data management
- Reduce administrative workload and paperwork
- Provide quick, role-based access to student records
- Improve data accuracy and security
- Generate reports and analytics for decision-making

3. Scope of the Project

Module	Description
User Authentication	Login with role-based access (Admin, Teacher, Student)
Student Management	Add, edit, delete, and view student details
Course Management	Manage courses, subjects, and class sections
Attendance Management	Record and track daily attendance
Marks/Result Management	Enter marks and generate report cards
Teacher Management	Manage teacher profiles and subject assignments
Fee Management	Track fee payments and receipts
Reports & Analytics	Student lists, performance reports, summaries
Dashboard	Role-specific dashboards with statistics

4. Target Users

Administrator – Full access to all modules

Teacher – Manage attendance, marks, view student details

Student – View profile, attendance, and results

Parent (optional) – View child's academic progress

5. System Modules and Features

The system is divided into the modules below. Each module handles one part of the work, so it is easy to build and test one by one.

5.1 User Authentication and Security

- Login and logout with a username and password

- Role-based access for Admin, Teacher, Student and Parent
- Passwords stored safely using PHP password_hash()
- Prepared statements to stop SQL injection, and input checking on every form

5.2 Student Management

- Add, edit, delete and view student details
- Search and filter students by name, class or ID
- Student profile page with attendance, marks and fee status

5.3 Teacher and Course Management

- Manage teacher profiles and assign subjects to teachers
- Create courses, subjects and class sections
- Link students to their courses and classes

5.4 Attendance Management

- Teachers record daily attendance (Present, Absent, Late)
- Students can view their own attendance record
- Attendance percentage and monthly summary

5.5 Marks and Result Management

- Enter marks for each subject and exam type (Test, Midterm, Final)
- Automatic total and grade calculation
- Printable report card for each student

5.6 Fee Management

- Record fee payments with amount and payment date
- Show paid and pending fees for each student
- Print fee receipts

5.7 Reports and Dashboard

- Role-specific dashboard with simple statistics
- Student lists, performance reports and attendance reports
- Fee collection and pending fee reports

6. Hardware and Software Requirements

6.1 Hardware Requirements

Item	Requirement
Processor	Dual-core CPU (Intel Core i3 or higher)
RAM	4 GB minimum (8 GB recommended)
Storage	10 GB free disk space
Other	Monitor, keyboard and mouse

6.2 Software Requirements

Item	Requirement
Operating System	Windows 10/11, Linux or macOS
Web Server	Apache (using XAMPP, WAMP or LAMP)
Database	MySQL Server 5.7 or higher (or MariaDB)
Language	PHP 7.4 or higher
Web Browser	Latest Chrome, Firefox or Edge
Tools	VS Code, phpMyAdmin, Git (optional)

7. Technology Stack

The project uses common, free tools that are easy to learn and work well together.

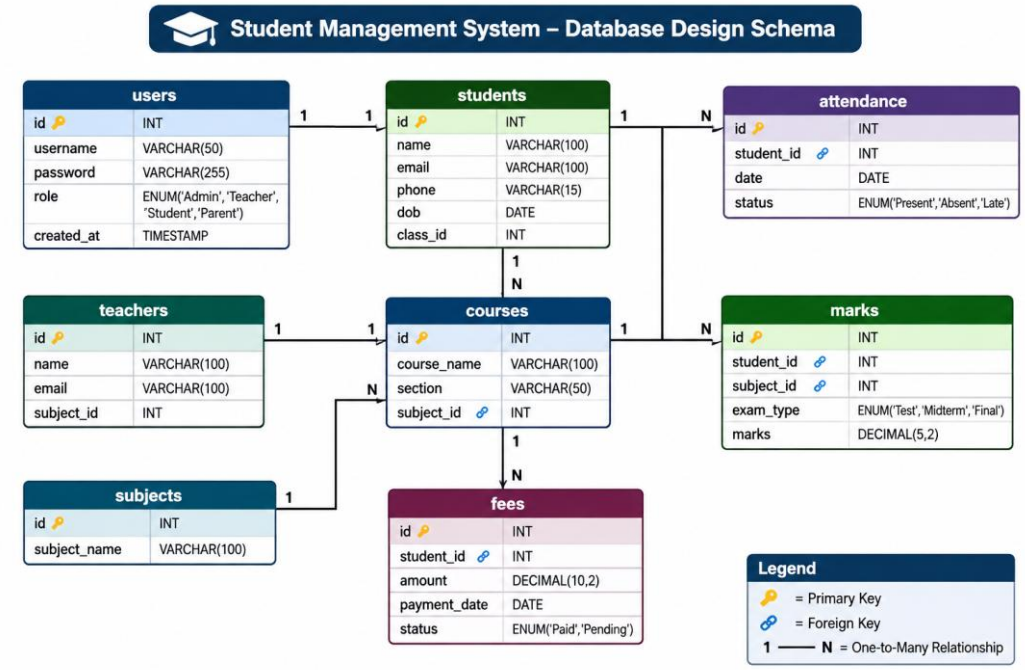
Layer	Technology	Used For
Frontend	HTML5, CSS3, JavaScript, Bootstrap	Pages, forms and responsive design
Backend	PHP (7.4+)	Business logic, sessions and validation
Database	MySQL	Storing all student records
Server	Apache (XAMPP/WAMP/LAMP)	Running the PHP application
Tools	VS Code, phpMyAdmin	Writing code and managing the database
Version Control	Git (optional)	Tracking changes in the code

8. Database Design (Overview)

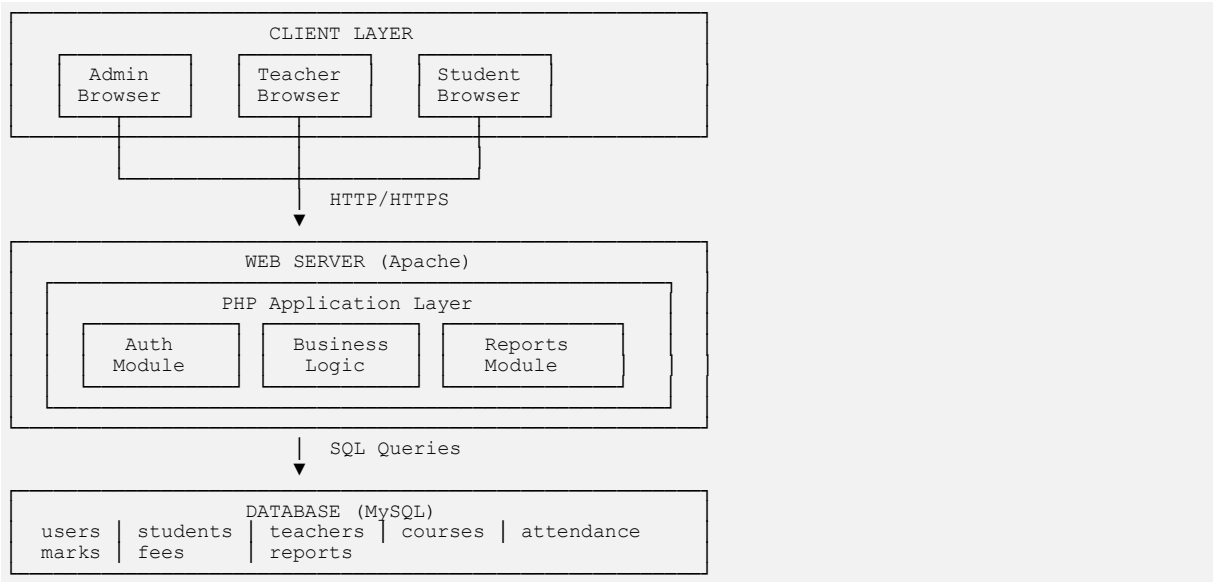
Key tables:

Table	Fields
users	id, username, password, role
students	id, name, email, phone, DOB, class_id
teachers	id, name, email, subject_id
courses	id, course_name, section
attendance	id, student_id, date, status
marks	id, student_id, subject_id, exam_type, marks
fees	id, student_id, amount, payment_date, status

Database Design Schema



9. System Architecture Diagram



10. Module Distribution

Module Workload Distribution

Module	% of Workload	Distribution
Student Management	25%	
Marks/Results	20%	

Module	% of Workload	Distribution
Attendance	15%	
Fees	15%	
Reports	15%	
Auth + Dashboard	10%	

11. Expected Benefits

Benefit Impact Analysis (Scale 1–10)

Benefit	Score	Impact (Scale 1–10)
Reduce manual work	10	
Faster report access	9	
Data accuracy	10	
Security	8	
Cost savings	7	
User satisfaction	8.5	

12. Risk Assessment (Matrix)

Likelihood ↓ / Impact →	Low	Medium	High
High		Data Loss	Server Downtime
Medium	Scope Creep	Security Breach	
Low			

Risk	Mitigation
Data loss	Regular backups
Security breach	Password hashing, prepared statements
Scope creep	Strict adherence to defined modules
Server downtime	Local dev server, cloud hosting for production

13. Conclusion

The Student Management System is a practical, scalable solution built on the proven PHP–MySQL stack. It offers a cost-effective, secure, and efficient way to manage student information while serving as an excellent demonstration of full-stack web development skills.

14. References

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